

FAR APART: THE UK AND PACIFIC TRADE

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FAR APART: THE UK AND PACIFIC TRADE - (2025, UNPUBLISHED): CONTRIBUTION

- ▶ It is (to the best of my knowledge) the first paper to develop a model a la CP to analyze the entrance of Uk into the CPTPP
- ▶ Most models have tariffs and non tariffs cost reduction but do not consider FTAs as hedging and strategical instruments
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INTRODUCTION

- ▶ The banking industry is highly concentrated. During the last 30 years the number of banks has fallen in half and the Top 10 share of assets has doubled.

MAIN QUESTION

- ▶ How much do policies enacted following Dodd-Frank (in particular, capital and liquidity requirements) affect failure rates and market shares of big and small banks in the U.S.?

ANSWER

- ▶ C-D find that higher capital requirements can have an important impact on the banking industry market structure
 - ▶ Leads to higher concentration due to a reduction in lending by small banks
 - ▶ This selection effect increases the cost of credit but improves allocative efficiency
 - ▶ Small welfare gains (losses) for households (entrepreneurs), with aggregate on order of Lucas' calculations for welfare costs of business cycles

OUTLINE

1. Data: Document U.S. Banking Facts from Balance sheet and Income Statement Panel Data.
2. Model:
 - ▶ Underlying static Cournot banking model with exogenous bank size distribution is from Allen & Gale (2004), Boyd & De Nicolo (2005)).
 - ▶ Endogenize bank size distribution by adding shocks and dynamic entry/exit decisions. Solve for industry equilibrium along the lines of Ericson & Pakes (1995) and Gowrisankaran & Holmes (2004).
 - ▶ Calibrate parameters to match long-run industry averages.
 - ▶ Test model against other moments: (1) business cycle correlations, and (2) the bank lending channel.
3. Policy Counterfactuals:
 - ▶ Decompose results of changes in capital and liquidity requirements.
 - ▶ Compare sensitivity between imperfect and perfect competition.

OLD MEMORIES

- ▶ We have already seen the C-P model in the lectures and in the PS hence we just refresh it.
- ▶ Why is it important? Add I-O linkages and sectors, but fundamentally is an EK. It is not a black box **CGE** model.
- ▶ **3 types of agents:**
 1. Consumers: $u(C_n) = \prod_{j=1}^J C_n^{j\alpha_n^j}$, where $\sum_{j=1}^J \alpha_n^j = 1$ with income I_n derived from work (ω_n) and rebate of tariffs.
 2. Intermediate Goods: $q_n^j(\omega^j) = z_n^j(\omega^j) [l_n^j(\omega^j)]^{\gamma_n^j} \prod_{k=1}^J [m_n^{k,j}(\omega^j)]^{\gamma_n^{k,j}}$, $\sum_{k=1}^J \gamma_n^{k,j} + \gamma_n^j = 1$
 3. Composite Intermediate Producers: $Q_n^j = \left[\int r_n^j(\omega^j)^{1-1/\sigma^j} d\omega^j \right]^{\sigma^j/(\sigma^j-1)}$
- ▶ input bundle cost is: $c_n^j = \Upsilon_n^j w_n^{\gamma_n^j} \prod_{k=1}^J P_n^k \gamma_n^{k,j}$, $\Upsilon_n^j \equiv \prod_{k=1}^J \left(\gamma_n^{k,j} \right)^{-\gamma_n^{k,j}} \left(\gamma_n^j \right)^{-\gamma_n^j}$ (1)
- ▶ **Where do we have I-O linkages?** [HINT:] $Q_n^j = C_n^j + \sum_{k=1}^J \int m_n^{j,k}(\omega^k) d\omega^k$.

EK WITH A TWIST...MORE ALGEBRA

Recall that pricing was in the form of $p_{n,i}^j(z_i^j) = \frac{c_i^j \kappa_{n,i}^j}{z_i^j}$. Indeed it's immediate to get that:

$$\text{Prob} \left[p_{n,i}^j(z_i^j) \leq p \right] = 1 - e^{-T_{ni}^j p^{\theta_j}}$$

Where $T_{ni}^j = \lambda_i^j \left(c_i^j \kappa_{ni}^j \right)^{-\theta_j}$. What is the distribution of the minimum price when a country goes shopping?

[HINT: $\text{Prob} \left[p_n^j \leq \min_{i \in I} p_{n,i}^j \right] = 1 - \prod_{i \in I} \text{prob}(p_{n,i}^j)$. Under what Assumption?] Hence we get that:

$$\text{Prob} \left[p_n^j \leq p \right] = 1 - e^{-\Phi_n^j p^{\theta_j}}$$

Where clearly $\Phi_n^j = \sum_{i=1}^N T_{ni}^j = \sum_{i=1}^N \lambda_i^j \left(c_i^j \kappa_{ni}^j \right)^{-\theta_j}$

What happens in non tradable sectors $\kappa_{i,n}^j = \infty$?

EK WITH A TWIST...MORE ALGEBRA II

Theorem

$p_n^j(\omega^j)^{\theta^j}$ has an exponential distribution.

Proof.

Define $g(x) = x^{\theta^j}$ where x has a Frechet distribution with parameters Φ_n^j and shape θ^j hence $f_x(x) = \theta^j \Phi_n^j x^{\theta^j-1} e^{-\Phi_n^j x^{\theta^j}}$. It follows that the density function of $y = g(x)$ is given by $f_y(y) = f_x(g^{-1}(y)) \left| \frac{\partial g^{-1}(y)}{\partial y} \right|$. Then since $g^{-1}(y) = y^{\frac{1}{\theta^j}}$, and $\frac{\partial g^{-1}(y)}{\partial y} = \frac{1}{\theta^j} y^{\frac{1}{\theta^j}-1}$, the density function of y is $f_y(y) = \Phi_n^j e^{-\Phi_n^j y}$, which is an exponential distribution with parameter Φ_n^j . $\square$

Now it's immediate to claim that:

$$(P_n^j)^{1-\eta^j} = \int \Phi_n^j y^{(1-\eta^j)/\theta^j} e^{-y\Phi_n^j} dy$$

EK WITH A TWIST...MORE ALGEBRA III

Now consider the change of variable $u = \Phi_n^j y$ then we get that:

$$(P_n^j)^{1-\sigma^j} = (\Phi_n^j)^{-(1-\sigma^j)/\theta^j} \int u^{(1-\sigma^j)/\theta^j} e^{-u} du$$

[HINT: Recall that $\Gamma(a) = \int_0^\infty u^{a-1} e^{-u} du$]

From which it should follow immediately that:

$$\text{Tradable : } P_n^j = A^j \Phi_n^{j-1/\theta^j} \quad (2)$$

$$\text{Non Tradable : } P_n^j = A^j \lambda_n^j (c_n^j)^{-\theta^j}$$

Where $A^j = \Gamma(\xi^j)^{1/(1-\sigma^j)}$, and $\Gamma(\xi^j)$ is a Gamma function evaluated at $\xi^j = 1 + (1 - \sigma^j) / \theta^j$.

EK WITH A TWIST...MORE ALGEBRA IV

Again we can derive the expenditure shares $\pi_{n,i}^j = X_{n,i}^j / X_n^j$ as:

$$X_{ni}^j = \Pr \left[\frac{c_i^j \kappa_{ni}^j}{z_i^j(\omega^j)} \leq \min_{h \neq i} \frac{c_h^j \kappa_{nh}^j}{z_h^j(\omega^j)} \right] X_n^j$$

[HINT: Recall that we have CES Preferences otherwise $X_n^j = \sum_{k=1}^J \xi_n^{j,k} \sum_{i=1}^N X_i^k \frac{\pi_{in}^k}{1 + \tau_{in}^k} + \alpha_n^j l_n$,]
From which is easy to get that:

$$X_{ni}^j = \frac{T_{ni}^j}{\Phi_n^j} \left[\int_0^\infty \Phi_n^j e^{-\Phi_n^j p^{\theta^j}} \theta^j p^{\theta^j-1} dp \right] X_n^j = \frac{\lambda_i^j \left(c_i^j \kappa_{ni}^j \right)^{-\theta^j}}{\sum_{i=1}^N \lambda_i^j \left(c_i^j \kappa_{ni}^j \right)^{-\theta^j}} X_n^j, \quad (3)$$

Recall in fact the law of minima for independent draws $\pi_{ni}^j = \int_0^\infty \prod_{s \neq i} \left[1 - G_{ns}^j(p) \right] dG_{ni}^j(p)$

ANOTHER TWIST: TRADE DEFICITS

We get that:

$$X_n^j = \sum_{k=1}^J \gamma_n^{j,k} \sum_{i=1}^N X_i^k \frac{\pi_{in}^k}{1 + \tau_{in}^k} + \alpha_n^j I_n, \quad (4)$$

$$I_n = w_n L_n + R_n + D_n,$$

Then using the definition of expenditure and trade deficit we get that:

$$\sum_{j=1}^J \sum_{i=1}^N X_n^j \frac{\pi_{ni}^j}{1 + \tau_{ni}^j} - D_n = \sum_{j=1}^J \sum_{i=1}^N X_i^j \frac{\pi_{in}^j}{1 + \tau_{in}^j} \quad (5)$$

Definition

Given L_n, D_n, λ_n^j and d_{ni}^j , an equilibrium under tariff structure τ is a wage vector $\mathbf{w} \in \mathbf{R}_{++}^N$ and prices $\left\{ P_n^j \right\}_{j=1, n=1}^{J, N}$ that satisfy equilibrium conditions (1), (2), (3), (4), and (5) for all j, n .

LOG LINEARIZE OR RELATIVE CHANGES?

- ▶ In macro we usually log linearize models hence we taylor approximate and then study the problem locally. This implies looking at % variations locally. E.g. $1 = R_{t+1}\beta (C_{t+1}/C_t)^{-\gamma}$ becomes

$$c_t \simeq -\sigma r_{t+1} + c_{t+1}$$

- ▶ So working on % changes (locally and around the ss) is something deeply diffused in DSGE.
- ▶ Do we need to do something similar even in our simplified CGE? While in macro we have a necessity to solve highly non linear and dynamic equations, in our model it's not about computational complexity. As a side effect calibration and IRFs is easier BUT comes with a great cost: losing second order effects (HANKS literature).
- ▶ **IDEA**: take the benefits (calibration, IRFs, loss of difficult to estimate parameters) and leave the costs (first order approximation).

LOG LINEARIZE OR RELATIVE CHANGES? RELATIVE CHANGES

- ▶ Take one of the first equations we need in our model under tariff structure (τ):

$$P_n^j = A^j \left[\sum_{i=1}^N \lambda_i^j \left(c_i^j \kappa_{ni}^j \right)^{-\theta^j} \right]^{-1/\theta^j}$$

- ▶ Now consider the same equation under tariff structure (τ'):

$$P_n^{j'} = A^j \left[\sum_{i=1}^N \lambda_i^j \left(c_i^{j'} \kappa_{ni}^{j'} \right)^{-\theta^j} \right]^{-1/\theta^j}$$

Taking the ratio we get that:

$$\frac{P_n^j}{P_n^{j'}} = \left[\sum_{i=1}^N \lambda_i^j \left(c_i^j \kappa_{ni}^j \right)^{-\theta^j} \right]^{-1/\theta^j}$$

$$\hat{c}_n^j = \hat{w}_n^{\gamma_n^j} \prod_{k=1}^J \hat{\rho}_n^k \gamma_n^{k,j}. \quad (6)$$

$$\hat{\rho}_n^j = \left[\sum_{i=1}^N \pi_{ni}^j \left[\hat{\kappa}_{ni}^j \hat{c}_i^j \right]^{-\theta^j} \right]^{-1/\theta^j} \quad (7)$$

$$\hat{\pi}_{ni}^j = \left[\frac{\hat{c}_i^j \hat{\kappa}_{ni}^j}{\hat{\rho}_n^j} \right]^{-\theta^j} \quad (8)$$

$$X_n^{j'} = \sum_{k=1}^J \gamma_n^{j,k} \sum_{i=1}^N \frac{\pi_{in}^{k'}}{1 + \tau_{in}^{k'}} X_i^{k'} + \alpha_n^j I_n' \quad (9)$$

$$\sum_{j=1}^J \sum_{i=1}^N \frac{\pi_{ni}^{j'}}{1 + \tau_{ni}^{j'}} X_n^{j'} - D_n = \sum_{j=1}^J \sum_{i=1}^N \frac{\pi_{in}^{j'}}{1 + \tau_{in}^{j'}} X_i^{j'} \quad (10)$$

WE KNOW, BUT WE DON'T

- ▶ During lectures we gave the intuitions on how to solve it, namely:
 1. guess a $\hat{w} = 1$.
 2. Combine (6) and (7). This is a $N \times J$ system in $N \times J$ unknowns. We get that we can find $\hat{p}_n(\hat{w})$ and hence $\hat{c}_n^j(\hat{w})$ (conditional on our guess)
 3. Use the information on π_{ni}^j and θ^j together with the solutions to $\hat{p}_n^j(\hat{w})$ and $\hat{c}_n^j(\hat{w})$ from step 2 and solve for $\pi_{ni}^{j'}(\hat{w})$ using (8).
 4. Given $\pi_{ni}^{j'}(\hat{w})$, τ' , γ_n^j , $\gamma_n^{j,k}$ and α_n^j , solve for total expenditure in each sector j and country n , $X_n^{j'}(\hat{w})$ consistent with the vector of wages $\hat{w}$ using (9).
 5. Substituting $\pi_{ni}^{j'}(\hat{w})$, $X_n^{j'}(\hat{w})$, τ' , and D_n into (10) verify if the trade balance holds.
 6. If not, adjust guess of $\hat{w}$ until equilibrium condition is obtained.
- ▶ How do we do step 4? Who is $I_n^{j'}$? Is it tedious to ask if $x_{nj'}$ depend just on $\hat{w}$?
- ▶ How do we adjust the guess ensuring convergence?

SMALL WORK

Note that we can re write eq. (9) as:

$$X_n^{j'} = \sum_{k=1}^J \gamma_n^{j,k} \sum_{i=1}^N \frac{\pi_{in}^k(\hat{\mathbf{w}})}{1 + \tau_{in}^{k'}} X_i^{k'} + \alpha_n^j \left(\hat{w}_n w_n L_n + \sum_{j=1}^J \sum_{i=1}^N \tau_{ni}^{j'} M_{ni}^{j'}(\hat{\mathbf{w}}) + D'_n \right).$$

Or in matrix form:

$$\mathbf{\Omega}(\hat{\mathbf{w}}) \mathbf{X} = \mathbf{\Delta}(\hat{\mathbf{w}}) \longleftrightarrow \mathbf{X}(\hat{\mathbf{w}}) = \mathbf{\Omega}^{-1}(\hat{\mathbf{w}}) \mathbf{\Delta}(\hat{\mathbf{w}})$$

- ▶ This is the **TRICK**! We have equilibrium conditions transformed as functions of our guess!
- ▶ how do we ensure convergence? Use a newton like like $w_n^{(t+1)} = w_n^{(t)} \cdot \left(1 - v \cdot \frac{S'_n}{VA_n}\right)$ works.

DATA

DATA

INSERIRE LA DESCRIZIONE DEI DATI
COME DA PAG1 READ ME (excl. T.txt)

WHAT IS WELFARE?

- ▶ in this model changes of real wages are **NOT** changes in welfare.
- ▶ To study the effect of changes in the counterfactual we need to study the effect of $W_n = \frac{I_n}{P_n}$ where we already encountered I_n while P_n is the consumption price index given by $P_n = \prod_{j=1}^J \left(P_n^j / \alpha_n^j \right)^{\alpha_n^j}$. Now totally differentiating we get that:

$$d \ln W_n = \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \underbrace{\left(E_{ni}^j d \ln c_n^j - M_{ni}^j d \ln c_i^j \right)}_{\text{Terms of trade}} + \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \underbrace{\tau_{ni}^j M_{ni}^j \left(d \ln M_{ni}^j - d \ln c_i^j \right)}_{\text{Volume of trade}},$$

- ▶ Algebra is becoming a bit too much and it's not the focus but if we want to indulge... bureaucratic fever

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WELFARE COMPUTATIONS [Back](#)

Some computations